

Roll No.....

Odd End Semester Examination December-2024

Name of the Program: B. Tech (ECE)
Name of the Course: Wireless Sensor Networks

Semester: VII
Course Code: TEC 755

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	CO1
(a)	Draw the architecture of a sensor node. List various modes of a sensor node.	
(b)	Explain in details the elements of WSN with the help of diagram.	
(c)	Write a short note on classification of Wireless Sensor Networks.	
Q2	(20 marks)	CO2
(a)	Derive the expression of detection advantage of a multi-hop network over a single-hop network.	
(b)		
(c)	Differentiate between flat and hierarchical architecture of WSN with the help of diagram.	
Q3	(20 marks)	CO3
(a)	Discuss about the Contention Based MAC Protocols with Scheduling Mechanisms.	
(b)	Discuss the main responsibilities of physical layer. Explain about the design parameters of physical layer.	
(c)	Write short notes on following: i. S-MAC protocol ii. IEEE 802.15.4 standard	
Q4	(20 marks)	CO4
(a)	List the classification of routing protocols. Explain any two in detail.	
(b)	What is geographical routing? Explain in detail.	
(c)	Discuss design issues of transport control protocols. Differentiate between TCP and UDP.	
Q5	(20 marks)	CO5 & CO6
(a)	How and why WSNs to Internet communication process is advantageous?	
(b)	What is the Gateway concept in WSN? Why gateways are required in WSNs?	
(c)	What are the hardware components of WSNs? Explain each one in detail.	

